SCENIC

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Loading packages

```
suppressMessages(library(dplyr))
suppressMessages(library(Seurat))
suppressMessages(library(patchwork))
suppressMessages(library(ggplot2))
suppressMessages(library(SCENIC))
suppressMessages(library(ComplexHeatmap))
suppressMessages(library(AUCell))
suppressMessages(library(formatR))
```

Loading data

```
Cell_10xAnnotate <- readRDS("../2-Paper/lavage_10x_annotated.rds")
```

prepare cellinfo et expression matrix

```
exprMat <- Cell_10xAnnotate@assays$RNA@data
cellInfo <- data.frame(seuratCluster=Idents(Cell_10xAnnotate))
```

```
cellInfo$Annotation <- Cell_10xAnnotate$Annotation
cellInfo$nGene <- colSums(exprMat>0)

dir.create ("int")
saveRDS(exprMat, file = "int/exprMat.Rds")
saveRDS(cellInfo, file = "int/cellInfo.Rds")
```

```
exprMat <- readRDS("int/exprMat.Rds")
cellInfo <- readRDS("int/cellInfo.Rds")
```

Download binding motifs for HUMAN

```
dbFiles <-
c("https://resources.aertslab.org/cistarget/databases/old/homo_sapiens/hg19/refseq_r45/mc9nr/gene_based/
"https://resources.aertslab.org/cistarget/databases/old/homo_sapiens/hg19/refseq_r45/mc9nr/gene_based/hg

dir.create("cisTarget_databases");
setwd("cisTarget_databases")
for(featherURL in dbFiles)
{
  download.file(featherURL, destfile=basename(featherURL)) # saved in current dir
}
```

Initialise Scenic

```
data(list="motifAnnotations_hgnc_v9", package="RcisTarget")
motifAnnotations_hgnc <- motifAnnotations_hgnc_v9

org <- "hgnc"
dbDir <- "cisTarget_databases"
dbDir <- path.expand(dbDir)
myDatasetTitle <- "SCENIC Analysis"
data(defaultDbNames)
dbs <- defaultDbNames[[org]]
scenicOptions <- initializeScenic(org=org, dbDir=dbDir, dbs=dbs, nCores=25)
```

```
## Motif databases selected:
##   hg19-500bp-upstream-7species.mc9nr.feather
##   hg19-tss-centered-10kb-7species.mc9nr.feather
```

```
## Using the column 'features' as feature index for the ranking database.
## Using the column 'features' as feature index for the ranking database.
```

Gene filtering

1. Filter by the total number of reads per gene. Keeps only the genes with at least 6 UMI counts across all samples.
2. Filter by the number of cells in which the gene is detected.

```r
exprMat <- as.matrix(exprMat) # need a regulat matrix
genesKept <- geneFiltering(exprMat, scenicOptions=scenicOptions,
                           minCountsPerGene=3*.01*ncol(exprMat),
                           minSamples=ncol(exprMat)*.01)

dim(exprMat) # 14386
exprMat_filtered <- exprMat[genesKept, ]
dim(exprMat_filtered) # 9389
```

Correlation

```r
runCorrelation(exprMat_filtered, scenicOptions)
```

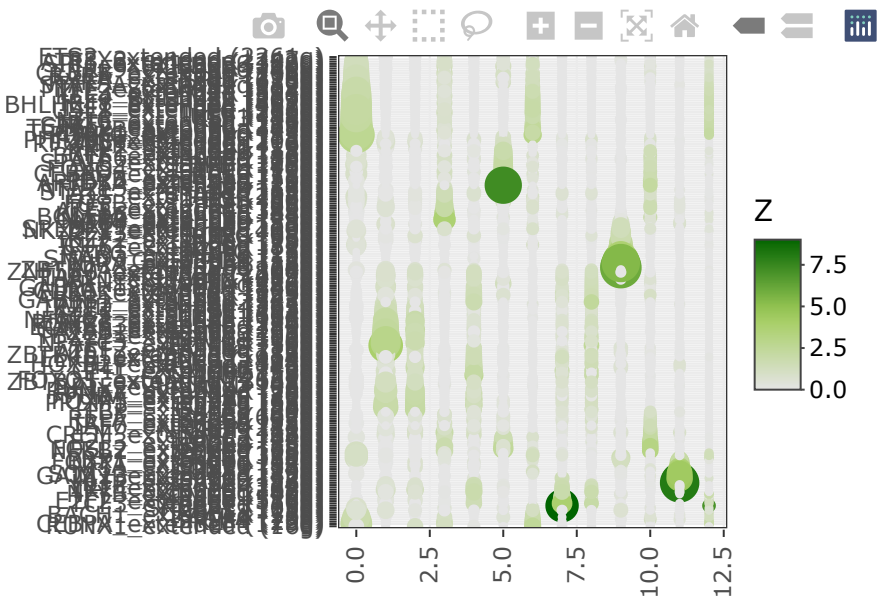
```r
runGenie3(exprMat_filtered, scenicOptions)
```

```r
scenicOptions@settings$verbose <- TRUE
scenicOptions@settings$nCores <- 10
scenicOptions@settings$seed <- 123

runSCENIC_1_coexNetwork2modules(scenicOptions)
runSCENIC_2_createRegulons(scenicOptions)
runSCENIC_3_scoreCells(scenicOptions, exprMat_filtered)
```

Plotting the results

Viewing markers based on specificity score

```r
regulonAUC <- loadInt(scenicOptions, "aucell_regulonAUC") # require file
int/3.4_regulonAUC.Rds


rss <- calcRSS(AUC=getAUC(regulonAUC), cellAnnotation=cellInfo[colnames(regulonAUC),
"seuratCluster"])
#rssPlot <- plotRSS(rss["Mafb (44g)",])
rssPlot <- plotRSS(rss)
plotly::ggplotly(rssPlot$plot)
```

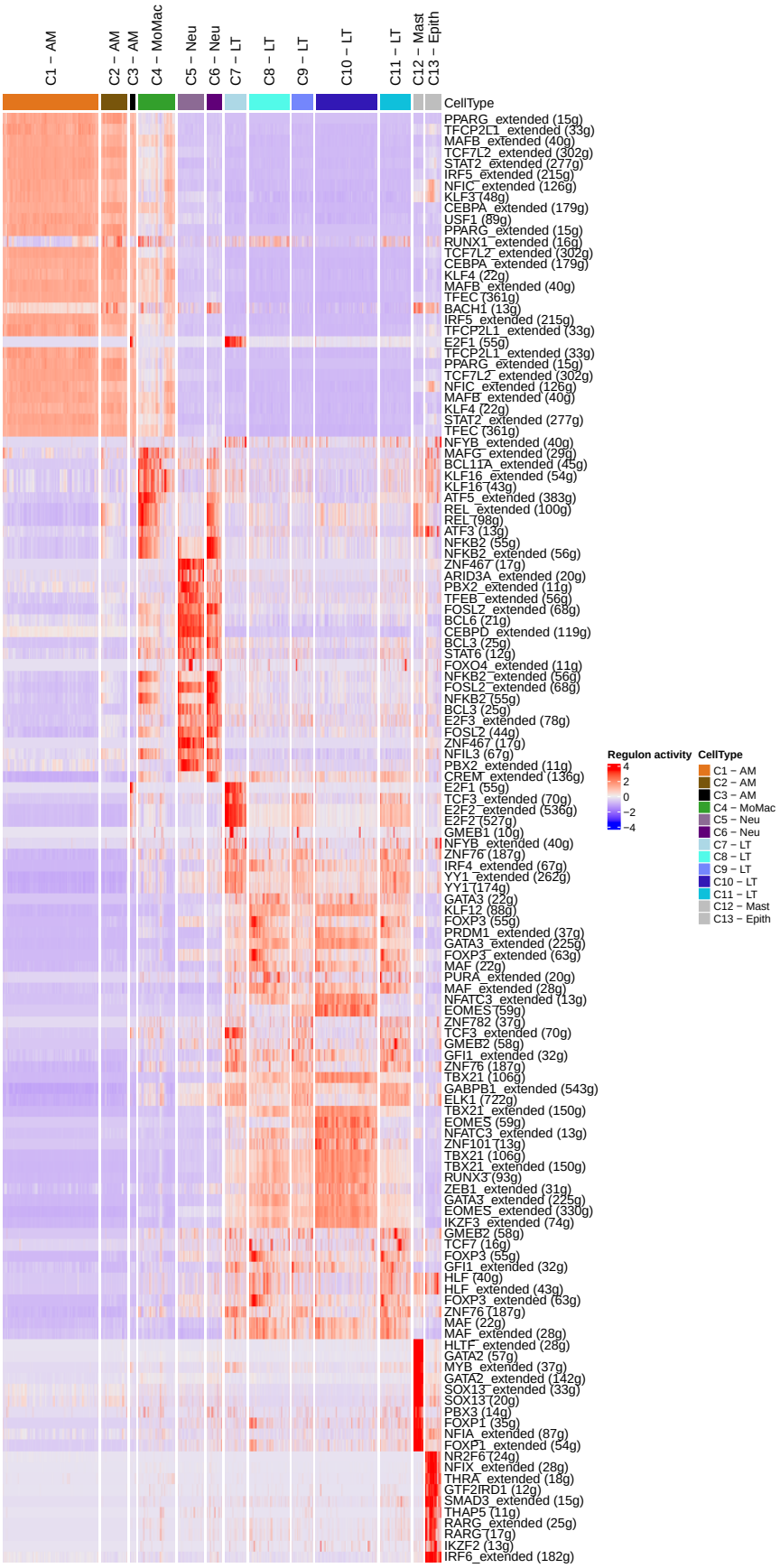


Plot on Heatmap the 10 markers the most specific of each cell type

```
#rss <- rss[,c(8,11,2,3,7,1,12,9,4,6,5,10)]
rss <- rss[,c(1,6,8,3,7,2,13,12,5,4,9,11,10)]

rss <- rss[,c(1,7,13,4,6,11,8,3,9,2,5,12,10)]

top_tf <- c()
for (i in 1:length(colnames(rss))) {

  TFs <- sort(rss[,i], decreasing = T)
  top_10 <- head(TFs, 10)
  top_tf <- c(top_tf,top_10)
}

length(top_tf)

#top_tf

# Creation of the Heatmap

regulonAUC.mat <- regulonAUC@assays@data@listData$AUC
Subset_regulonActivity <-regulonAUC.mat[names(top_tf),]

regulonActivity_byCellType_Scaled <- t(scale(t(Subset_regulonActivity), center = T,
scale=T))

colors <- c("#E3751C", "#3218b5", "#52fae9", "#33A02C", "#0fc0db", "#8c6a94", "#78550b",
"lightblue", "#7486fc", "grey", "#600078", "gray", "black")

names(colors) <- levels(Cell_10xAnnotate$Annotation)

df <- as.data.frame(Cell_10xAnnotate$Annotation)
colnames(df) <- "CellType"
```

```r
color_df <- list(CellType = c("C1 - AM" = "#E3751C",
                              "C2 - AM" = "#78550b",
                              "C3 - AM" = "black",
                              "C4 - MoMac" = "#33A02C",
                              "C5 - Neu" = "#8c6a94",
                              "C6 - Neu" = "#600078",
                              "C7 - LT" = "lightblue",
                              "C8 - LT" = "#52fae9",
                              "C9 - LT" = "#7486fc",
                              "C10 - LT" = "#3218b5",
                              "C11 - LT" = "#0fc0db",
                              "C12 - Mast" = "grey",
                              "C13 - Epith" = "grey"))

Heatmap <- Heatmap(regulonActivity_byCellType_Scaled, name="Regulon activity",
show_column_names = FALSE,
        column_split = factor(Cell_10xAnnotate$Annotation),
        cluster_column_slices = F,
        cluster_rows = F,
        top_annotation = HeatmapAnnotation(df = df, col = color_df),
        show_column_dend = F,
        column_title_rot = 90)

#tidyHeatmap::save_pdf(Heatmap,"Scenic_10x.pdf", width = 40, height = 50, units = "cm")
```

```r
Heatmap
```


C1 – AM
C2 – AM
C3 – AM
C4 – MoMac
C5 – Neu
C6 – Neu
C7 – LT
C8 – LT
C9 – LT
C10 – LT
C11 – LT
C12 – Mast
C13 – Epith
CellType
PPARG_extended (15g)
TFCP2L1_extended (33g)
MAFB_extended (40g)
TCF7L2_extended (302g)
STAT2_extended (277g)
IRF5_extended (215g)
NFIC_extended (126g)
KLF3 (48g)
CEBPA_extended (179g)
USF1 (89g)
PPARG_extended (15g)
RUNX1_extended (16g)
TCF7L2_extended (302g)
CEBPA_extended (179g)
KLF4 (22g)
MAFB_extended (40g)
TFEC (361g)
BACH1 (13g)
IRF5_extended (215g)
TFCP2L1_extended (33g)
E2F1 (55g)
TFCP2L1_extended (33g)
PPARG_extended (15g)
TCF7L2_extended (302g)
NFIC_extended (126g)
MAFB_extended (40g)
KLF4 (22g)
STAT2_extended (277g)
TFEC (361g)
NFYB_extended (40g)
MAFG_extended (29g)
BCL11A_extended (45g)
KLF16_extended (54g)
KLF16 (43g)
ATF5_extended (383g)
REL_extended (100g)
REL (98g)
ATF3 (13g)
NFKB2 (55g)
NFKB2_extended (56g)
ZNF467 (17g)
ARID3A_extended (20g)
PBX2_extended (11g)
TFEB_extended (56g)
FOSL2_extended (68g)
BCL6 (21g)
CEBPD_extended (119g)
BCL3 (25g)
STAT6 (12g)
FOXO4_extended (11g)
NFKB2_extended (56g)
FOSL2_extended (68g)
NFKB2 (55g)
BCL3 (25g)
E2F3_extended (78g)
FOSL2 (44g)
ZNF467 (17g)
NFIL3 (67g)
PBX2_extended (11g)
CREM_extended (136g)
E2F1 (55g)
TCF3_extended (70g)
E2F2_extended (536g)
E2F2 (527g)
GMEB1 (10g)
NFYB_extended (40g)
ZNF76 (187g)
IRF4_extended (67g)
YY1_extended (262g)
YY1 (174g)
GATA3 (22g)
KLF12 (88g)
FOXP3 (55g)
PRDM1_extended (37g)
GATA3_extended (225g)
FOXP3_extended (63g)
MAF (22g)
PURA_extended (20g)
MAF_extended (28g)
NFATC3_extended (13g)
EOMES (59g)
ZNF782 (37g)
TCF3_extended (70g)
GMEB2 (58g)
GFI1_extended (32g)
ZNF76 (187g)
TBX21 (106g)
GABPB1_extended (543g)
ELK1 (722g)
TBX21_extended (150g)
EOMES (59g)
NFATC3_extended (13g)
ZNF101 (13g)
TBX21 (106g)
TBX21_extended (150g)
RUNX3 (93g)
ZEB1_extended (31g)
GATA3_extended (225g)
EOMES_extended (330g)
IKZF3_extended (74g)
GMEB2 (58g)
TCF7 (16g)
FOXP3 (55g)
GFI1_extended (32g)
HLF (40g)
HLF_extended (43g)
FOXP3_extended (63g)
ZNF76 (187g)
MAF (22g)
MAF_extended (28g)
HLTF_extended (28g)
GATA2 (57g)
MYB_extended (37g)
GATA2_extended (142g)
SOX13_extended (33g)
SOX13 (20g)
PBX3 (14g)
FOXP1 (35g)
NFIA_extended (87g)
FOXP1_extended (54g)
NR2F6 (24g)
NFIX_extended (28g)
THRA_extended (18g)
GTF2IRD1 (12g)
SMAD3_extended (15g)
THAP5 (11g)
RARG_extended (25g)
RARG (17g)
IKZF2 (13g)
IRF6_extended (182g)
Regulon activity
4
2
0
−2
−4
CellType
C1 – AM
C2 – AM
C3 – AM
C4 – MoMac
C5 – Neu
C6 – Neu
C7 – LT
C8 – LT
C9 – LT
C10 – LT
C11 – LT
C12 – Mast
C13 – Epith

```r
sessionInfo()
```

```
## R version 4.3.2 (2023-10-31)
## Platform: x86_64-pc-linux-gnu (64-bit)
## Running under: Ubuntu 22.04.3 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.20.so;  LAPACK version 3.10.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=fr_BE.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=fr_BE.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=fr_BE.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=fr_BE.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Brussels
## tzcode source: system (glibc)
##
## attached base packages:
## [1] grid      stats     graphics  grDevices utils     datasets  methods
## [8] base
##
## other attached packages:
##  [1] data.table_1.14.8     BiocParallel_1.34.2   RcisTarget_1.20.0
##  [4] formatR_1.14          AUCell_1.22.0         ComplexHeatmap_2.16.0
##  [7] SCENIC_1.3.1          ggplot2_3.4.2         patchwork_1.1.2
## [10] SeuratObject_4.1.3    Seurat_4.3.0          dplyr_1.1.2
##
## loaded via a namespace (and not attached):
##   [1] RcppAnnoy_0.0.21            splines_4.3.2
##   [3] later_1.3.1                 bitops_1.0-7
##   [5] tibble_3.2.1                R.oo_1.25.0
##   [7] polyclip_1.10-4             graph_1.78.0
##   [9] XML_3.99-0.14               lifecycle_1.0.3
##  [11] doParallel_1.0.17           processx_3.8.1
##  [13] globals_0.16.2              lattice_0.22-5
##  [15] MASS_7.3-60                 crosstalk_1.2.0
##  [17] magrittr_2.0.3              plotly_4.10.2
##  [19] rmarkdown_2.23              yaml_2.3.7
##  [21] httpuv_1.6.11               sctransform_0.3.5
##  [23] spam_2.9-1                  sp_2.0-0
##  [25] spatstat.sparse_3.0-2       reticulate_1.30
##  [27] chromote_0.1.2              cowplot_1.1.1
##  [29] pbapply_1.7-2               DBI_1.1.3
##  [31] RColorBrewer_1.1-3          abind_1.4-5
##  [33] zlibbioc_1.46.0             Rtsne_0.16
##  [35] GenomicRanges_1.52.0        purrr_1.0.1
##  [37] R.utils_2.12.2              BiocGenerics_0.46.0
##  [39] RCurl_1.98-1.12             webshot2_0.1.1
##  [41] circlize_0.4.15             GenomeInfoDbData_1.2.10
```

```
##  [43] IRanges_2.34.0              S4Vectors_0.38.1
##  [45] ggrepel_0.9.3               irlba_2.3.5.1
##  [47] listenv_0.9.0               spatstat.utils_3.0-3
##  [49] goftest_1.2-3               spatstat.random_3.1-5
##  [51] annotate_1.78.0             fitdistrplus_1.1-11
##  [53] parallelly_1.36.0           DelayedMatrixStats_1.22.1
##  [55] leiden_0.4.3                codetools_0.2-19
##  [57] DelayedArray_0.26.3         shape_1.4.6
##  [59] tidyselect_1.2.0            farver_2.1.1
##  [61] matrixStats_1.0.0           stats4_4.3.2
##  [63] spatstat.explore_3.2-1      jsonlite_1.8.7
##  [65] GetoptLong_1.0.5            ellipsis_0.3.2
##  [67] progressr_0.13.0            iterators_1.0.14
##  [69] ggridges_0.5.4              survival_3.5-7
##  [71] foreach_1.5.2               tools_4.3.2
##  [73] ica_1.0-3                   Rcpp_1.0.11
##  [75] glue_1.6.2                  gridExtra_2.3
##  [77] xfun_0.39                   MatrixGenerics_1.12.2
##  [79] websocket_1.4.1             GenomeInfoDb_1.36.0
##  [81] withr_2.5.0                 fastmap_1.1.1
##  [83] fansi_1.0.4                 digest_0.6.33
##  [85] R6_2.5.1                    mime_0.12
##  [87] colorspace_2.1-0            scattermore_1.2
##  [89] tensor_1.5                  spatstat.data_3.0-1
##  [91] RSQLite_2.3.1               R.methodsS3_1.8.2
##  [93] utf8_1.2.3                  tidyr_1.3.0
##  [95] generics_0.1.3              httr_1.4.6
##  [97] htmlwidgets_1.6.2           S4Arrays_1.0.4
##  [99] uwot_0.1.16                 pkgconfig_2.0.3
## [101] gtable_0.3.3                blob_1.2.4
## [103] lmtest_0.9-40               XVector_0.40.0
## [105] htmltools_0.5.5             dotCall64_1.0-2
## [107] clue_0.3-64                 GSEABase_1.62.0
## [109] scales_1.2.1                Biobase_2.60.0
## [111] png_0.1-8                   knitr_1.43
## [113] rstudioapi_0.14             rjson_0.2.21
## [115] reshape2_1.4.4              nlme_3.1-163
## [117] GlobalOptions_0.1.2         zoo_1.8-12
## [119] cachem_1.0.8                stringr_1.5.0
## [121] KernSmooth_2.23-22          parallel_4.3.2
## [123] miniUI_0.1.1.1              arrow_12.0.1
## [125] AnnotationDbi_1.62.1        pillar_1.9.0
## [127] vctrs_0.6.3                 RANN_2.6.1
## [129] promises_1.2.0.1            xtable_1.8-4
## [131] cluster_2.1.4               evaluate_0.21
## [133] magick_2.7.5                cli_3.6.1
## [135] compiler_4.3.2              rlang_1.1.1
## [137] crayon_1.5.2                future.apply_1.11.0
## [139] labeling_0.4.2              ps_1.7.5
## [141] plyr_1.8.8                  stringi_1.7.12
## [143] viridisLite_0.4.2           deldir_1.0-9
## [145] assertthat_0.2.1            munsell_0.5.0
## [147] Biostrings_2.68.1           lazyeval_0.2.2
## [149] spatstat.geom_3.2-4         Matrix_1.6-1
```

```
## [151] sparseMatrixStats_1.12.0    bit64_4.0.5
## [153] future_1.33.0               KEGGREST_1.40.0
## [155] shiny_1.7.4.1               highr_0.10
## [157] SummarizedExperiment_1.30.2 ROCR_1.0-11
## [159] igraph_1.5.0.1              memoise_2.0.1
## [161] bit_4.0.5
```